

Chapter 20: 120 Beginner Practice Problems

Personal Notes — Practice Set

Logic building is a skill — and like every skill, it grows only with practice. This chapter is a curated set of 120 beginner problems covering everything from basic output to arrays, strings, and methods. The goal is to give you enough variety to stretch your thinking without overwhelming you.

Work through them at your own pace. Don't peek at solutions immediately — try, fail, dry-run on paper, and try again. That struggle is where real learning happens.

How to use this set: Solve each problem on paper first, then code it. After it works, ask yourself: "Is there a shorter way? A faster way?" That second pass is where most of the growth happens.

How the problems are organized

- Section 1 — Basics & Output (Q1–Q10)
- Section 2 — Variables & Data Types (Q11–Q18)
- Section 3 — Operators & Conditionals (Q19–Q30)
- Section 4 — Loops Basics (Q31–Q45)
- Section 5 — Number Logic & Maths (Q46–Q65)
- Section 6 — Patterns (Q66–Q80)
- Section 7 — 1D Arrays (Q81–Q95)
- Section 8 — 2D Arrays (Q96–Q103)
- Section 9 — Strings (Q104–Q115)
- Section 10 — Methods (Q116–Q120)

Section 1 — Basics & Output (Q1–Q10)

Get comfortable with `println`, basic syntax, and producing output.

Q1. Hello World

Problem: Print "Hello World" to the console.

Output: Hello World

Q2. Your Name

Problem: Print your name on the screen.

Output: Akanshu

Q3. Multi-line Greeting

Problem: Print three lines: Welcome / To / Java.

Output: Welcome To Java

Hint: Use 3 separate `println` calls

Q4. Print Numbers 1 to 5

Problem: Print numbers 1 to 5 on separate lines without a loop.

Output: 1 2 3 4 5

Hint: Use 5 println statements

Q5. Single Line Output

Problem: Print 1 to 5 on the SAME line, separated by space.

Output: 1 2 3 4 5

Hint: Use print, not println

Q6. Pi Value

Problem: Print the value of pi (3.14159).

Output: 3.14159

Q7. Box of Stars

Problem: Print 5 stars in a row on one line.

Output: * * * * *

Q8. Personal Card

Problem: Print your name, age, city on three lines.

Output: Aman 21 Mumbai

Q9. ASCII Art Smiley

Problem: Print a 3-line smiley face using : and)

Output: :) -- :~)

Q10. Print Without Newline

Problem: Print "Java" followed by "Rocks" on the same line.

Output: JavaRocks

Hint: Use print twice

Section 2 — Variables & Data Types (Q11–Q18)

Practice declaring variables of different types and printing them.

Q11. Sum of Two Numbers

Problem: Take two integers and print their sum.

Input: 5 7

Output: 12

Q12. Area of Rectangle

Problem: Given length and width, print the area.

Input: 4 6

Output: 24

Hint: area = length × width

Q13. Circle Area

Problem: Given radius, print the area of a circle.

Input: 5

Output: 78.5

Hint: Use $3.14 \times r \times r$

Q14. Swap Two Numbers

Problem: Take two integers and print them with values swapped.

Input: 3 7

Output: 7 3

Hint: Use a temp variable

Q15. Celsius to Fahrenheit

Problem: Convert Celsius to Fahrenheit. Formula: $F = (C \times 9/5) + 32$.

Input: 25

Output: 77.0

Hint: Use double for accuracy

Q16. Sum, Diff, Product

Problem: Print sum, difference, and product of two integers.

Input: 10 3

Output: 13 7 30

Q17. Average of Three Numbers

Problem: Take 3 numbers and print their average.

Input: 10 20 30

Output: 20.0

Hint: Use double for division

Q18. Total and Percentage

Problem: Take marks in 5 subjects (out of 100). Print total and percentage.

Input: 80 90 70 85 95

Output: Total: 420\nPercentage: 84.0

Section 3 — Operators & Conditionals (Q19–Q30)

Practice if-else, comparison operators, and decision-making.

Q19. Even or Odd

Problem: Check if a number is even or odd.

Input: 7

Output: Odd

Hint: $n \% 2 == 0 \rightarrow \text{even}$

Q20. Positive, Negative, Zero

Problem: Print whether a number is positive, negative, or zero.

Input: -5

Output: Negative

Q21. Largest of Two

Problem: Print the larger of two numbers.

Input: 10 7

Output: 10

Q22. Largest of Three

Problem: Print the largest of three numbers.

Input: 5 9 4

Output: 9

Hint: Use nested if or && operators

Q23. Smallest of Three

Problem: Print the smallest of three numbers.

Input: 5 9 4

Output: 4

Q24. Leap Year Check

Problem: Check if a year is a leap year.

Input: 2024

Output: Leap Year

Hint: Divisible by 4, but not 100 unless 400

Q25. Grade Calculator

Problem: Given marks, print Grade. ≥ 90 A, ≥ 80 B, ≥ 70 C, ≥ 60 D, else F.

Input: 85

Output: B

Hint: Use if-else-if ladder

Q26. Vowel or Consonant

Problem: Check if a single character is a vowel or consonant.

Input: e

Output: Vowel

Hint: Check against a, e, i, o, u

Q27. Eligibility to Vote

Problem: Take age. If ≥ 18 print "Eligible to vote", else "Not eligible."

Input: 20

Output: Eligible to vote

Q28. Three Numbers Equal

Problem: Check if three given numbers are all equal.

Input: 5 5 5

Output: All equal

Q29. Number in Range

Problem: Check if a number lies between 1 and 100 (inclusive).

Input: 55

Output: In range

Q30. Day of Week

Problem: Take a number 1-7 and print the corresponding day (1=Monday).

Input: 3

Output: Wednesday

Hint: Use switch or if-else

Section 4 — Loops Basics (Q31-Q45)

Use for and while loops to repeat actions and process series of values.

Q31. Print 1 to N

Problem: Print all numbers from 1 to N.

Input: 5

Output: 1 2 3 4 5

Hint: Simple for loop

Q32. Print N to 1

Problem: Print numbers from N down to 1.

Input: 5

Output: 5 4 3 2 1

Hint: Decrement loop

Q33. Print Even Numbers 1 to N

Problem: Print all even numbers from 1 to N.

Input: 10

Output: 2 4 6 8 10

Q34. Print Odd Numbers 1 to N

Problem: Print all odd numbers from 1 to N.

Input: 10

Output: 1 3 5 7 9

Q35. Sum of 1 to N

Problem: Find the sum of all numbers from 1 to N.

Input: 5

Output: 15

Hint: 1+2+3+4+5

Q36. Sum of Even Numbers

Problem: Find the sum of even numbers from 1 to N.

Input: 10

Output: 30

Hint: 2+4+6+8+10

Q37. Sum of Odd Numbers

Problem: Find the sum of odd numbers from 1 to N.

Input: 10

Output: 25

Q38. Multiplication Table

Problem: Print the multiplication table of N (1 to 10).

Input: 5

Output: 5 10 15 ... 50

Hint: Loop 1 to 10, multiply by N

Q39. Tables 1 to 10

Problem: Print the tables of 1 through 10 (each on its own block).

Output: Long output

Hint: Nested loop

Q40. Count Digits

Problem: Count the number of digits in a given number.

Input: 12345

Output: 5

Hint: Use / 10 in a loop

Q41. Sum of Digits

Problem: Find the sum of digits of a number.

Input: 1234

Output: 10

Hint: $n \% 10$ gives last digit

Q42. Reverse a Number

Problem: Print the digits of a number reversed.

Input: 1234

Output: 4321

Hint: $rev = rev * 10 + digit$

Q43. Power of a Number

Problem: Compute A^B (without Math.pow).

Input: 2 5

Output: 32

Hint: Multiply A by itself B times

Q44. Factorial

Problem: Compute factorial of N.

Input: 5

Output: 120

Hint: $5 \times 4 \times 3 \times 2 \times 1$

Q45. Print Fibonacci to N Terms

Problem: Print the first N Fibonacci numbers.

Input: 7

Output: 0 1 1 2 3 5 8

Hint: Each = sum of previous two

Section 5 — Number Logic & Maths (Q46–Q65)

These are the classic number-crunching problems that build real DSA thinking.

Q46. Prime Check

Problem: Check whether a number is prime.

Input: 7

Output: Prime

Hint: Loop 2 to $\sqrt{n}$

Q47. Primes 1 to N

Problem: Print all prime numbers between 1 and N.

Input: 20

Output: 2 3 5 7 11 13 17 19

Q48. Palindrome Number

Problem: Check if a number reads the same forward and backward.

Input: 121

Output: Palindrome

Hint: Reverse and compare

Q49. Armstrong Number (3-digit)

Problem: Check if a 3-digit number equals sum of cubes of its digits.

Input: 153

Output: Armstrong

Hint: $1^3 + 5^3 + 3^3$

Q50. Perfect Number

Problem: Check if a number equals the sum of its proper divisors.

Input: 28

Output: Perfect

Hint: $1 + 2 + 4 + 7 + 14 = 28$

Q51. GCD of Two Numbers

Problem: Find the GCD using Euclidean algorithm.

Input: 12 18

Output: 6

Hint: $\text{GCD}(a,b) = \text{GCD}(b, a\%b)$

Q52. LCM of Two Numbers

Problem: Find LCM using formula $(a*b)/\text{GCD}$.

Input: 12 18

Output: 36

Q53. Count Even Digits

Problem: Count how many digits in a number are even.

Input: 248531

Output: 3

Hint: 2, 4, 8 are even

Q54. Count Odd Digits

Problem: Count how many digits are odd.

Input: 248531

Output: 3

Hint: 5, 3, 1 are odd

Q55. Largest Digit

Problem: Find the largest digit in a number.

Input: 3729

Output: 9

Hint: Track max while extracting

Q56. Smallest Digit

Problem: Find the smallest digit in a number.

Input: 3729

Output: 2

Q57. Sum of Squares of Digits

Problem: Find sum of squares of all digits.

Input: 123

Output: 14

Hint: $1^2+2^2+3^2$

Q58. Strong Number

Problem: A number equal to sum of factorial of digits. ($145 = 1!+4!+5!$)

Input: 145

Output: Strong

Q59. Harshad Number

Problem: Number divisible by its digit sum. ($18 \rightarrow 1+8=9$, $18\%9==0$)

Input: 18

Output: Harshad

Q60. Happy Number

Problem: Replace number with sum of squares of digits repeatedly. Happy if reaches 1.

Input: 19

Output: Happy

Hint: Loop until 1 or detected cycle

Q61. Automorphic Number

Problem: Square ends with the number. ($5^2=25$, $25^2=625$, $76^2=5776$)

Input: 25

Output: Automorphic

Hint: Check last digits of square

Q62. Print Divisors

Problem: Print all divisors of N.

Input: 12

Output: 1 2 3 4 6 12

Q63. Sum of Divisors

Problem: Find sum of all divisors of N.

Input: 12

Output: 28

Q64. Count Divisors

Problem: Count how many divisors a number has.

Input: 12

Output: 6

Q65. Power of 2 Check

Problem: Check if a number is a power of 2.

Input: 16

Output: Yes

Hint: Keep dividing by 2, check if result is 1

Section 6 — Patterns (Q66–Q80)

Patterns are loop-mastery in disguise. Each one trains row-column thinking.

Q66. Solid Square

Problem: Print an N×N square of stars.

Input: 4

Output: 4 lines of 4 stars each

Q67. Hollow Square

Problem: Print a hollow square of stars (border only).

Input: 5

Output: Border stars, interior spaces

Q68. Right Triangle (Stars)

Problem: Print a right triangle with N rows; row i has i stars.

Input: 5

Output: Triangle growing right

Q69. Inverted Right Triangle

Problem: Triangle that shrinks each row.

Input: 5

Output: 5,4,3,2,1 stars

Hint: stars = n - row + 1

Q70. Solid Pyramid

Problem: Centered pyramid of N rows.

Input: 4

Output: Center-aligned stars

Hint: spaces = n-row, stars = 2*row-1

Q71. Inverted Pyramid

Problem: Pyramid upside-down.

Input: 4

Output: Top-wide, narrows down

Q72. Hollow Pyramid

Problem: Pyramid with only border stars.

Input: 5

Output: Border only

Q73. Solid Diamond

Problem: Pyramid + inverted pyramid combined.

Input: 4

Output: Diamond shape

Q74. Hollow Diamond

Problem: Diamond outline only.

Input: 4

Output: Border only diamond

Q75. Butterfly Pattern

Problem: Two triangles forming a butterfly shape.

Input: 4

Output: Symmetric butterfly

Q76. Number Triangle

Problem: Row i prints numbers 1 to i.

Input: 5

Output: 1\n1 2\n1 2 3 ...

Q77. Number Pyramid

Problem: Row i prints i repeated i times, centered.

Input: 4

Output: Centered pyramid of numbers

Q78. Floyd's Triangle

Problem: Triangle with consecutive integers filling each row.

Input: 5

Output: 1\n2 3\n4 5 6...

Q79. Alphabet Triangle

Problem: Row i prints A through (the i-th letter).

Input: 5

Output: A\nA B\nA B C ...

Q80. Pascal's Triangle

Problem: Each number = sum of two above it.

Input: 5

Output: 1\n1 1\n1 2 1\n1 3 3 1\n1 4 6 4 1

Hint: Use factorial or running product

Section 7 — 1D Arrays (Q81–Q95)

Practice with arrays — indices, traversal, searching, and basic manipulation.

Q81. Sum of Array

Problem: Find the sum of all elements in an array.

Input: [10, 20, 30, 40]

Output: 100

Q82. Average of Array

Problem: Find the average of array elements.

Input: [10, 20, 30, 40]

Output: 25.0

Q83. Maximum Element

Problem: Find the largest element in an array.

Input: [5, 9, 2, 8, 4]

Output: 9

Hint: Track max in a loop

Q84. Minimum Element

Problem: Find the smallest element in an array.

Input: [5, 9, 2, 8, 4]

Output: 2

Q85. Count Even & Odd

Problem: Count even and odd numbers in an array.

Input: [1, 2, 3, 4, 5]

Output: Even: 2, Odd: 3

Q86. Linear Search

Problem: Search for a target value. Return its index, or -1.

Input: [10,20,30] target=20

Output: 1

Q87. Reverse an Array

Problem: Reverse the array in place.

Input: [1, 2, 3, 4, 5]

Output: [5, 4, 3, 2, 1]

Hint: Two-pointer swap

Q88. Check Sorted

Problem: Check if the array is sorted in ascending order.

Input: [1, 3, 5, 7]

Output: Sorted

Q89. Second Largest

Problem: Find the second largest element in an array.

Input: [5, 9, 2, 8, 4]

Output: 8

Hint: Track largest and 2nd largest

Q90. Second Smallest

Problem: Find the second smallest element.

Input: [5, 9, 2, 8, 4]

Output: 4

Q91. Count Occurrences

Problem: Count how many times a value appears in the array.

Input: [1,2,3,2,4,2] x=2

Output: 3

Q92. Find Duplicate

Problem: Find a value that appears more than once.

Input: [1, 2, 3, 2, 5]

Output: 2

Hint: Nested loop or set

Q93. Move Zeros to End

Problem: Move all zeros to the end, keeping order of others.

Input: [0,1,0,3,12]

Output: [1,3,12,0,0]

Q94. Array Sum of Two

Problem: Given target, find two indices whose values add to target.

Input: [2,7,11,15] t=9

Output: 0 1

Hint: Nested loop, or use HashMap

Q95. Rotate Array Left by 1

Problem: Rotate the array one step left.

Input: [1,2,3,4,5]

Output: [2,3,4,5,1]

Hint: Save first, shift others

Section 8 — 2D Arrays (Q96–Q103)

Matrix problems — nested loops and row/column thinking.

Q96. Print Matrix

Problem: Print a 2D matrix in row-by-row format.

Input: [[1,2],[3,4]]

Output: 1 2\n3 4

Q97. Sum of Matrix

Problem: Find the sum of all elements in a 2D matrix.

Input: [[1,2],[3,4]]

Output: 10

Q98. Maximum in Matrix

Problem: Find the largest element in a matrix.

Input: [[1,5],[3,2]]

Output: 5

Q99. Row-wise Sum

Problem: Print sum of each row.

Input: [[1,2,3],[4,5,6]]

Output: Row 0: 6\nRow 1: 15

Q100. Column-wise Sum

Problem: Print sum of each column.

Input: [[1,2,3],[4,5,6]]

Output: Col 0: 5\nCol 1: 7\nCol 2: 9

Hint: Flip inner/outer loops

Q101. Diagonal Elements

Problem: Print the main diagonal of a square matrix.

Input: [[1,2,3],[4,5,6],[7,8,9]]

Output: 1 5 9

Hint: matrix[i][i]

Q102. Diagonal Sum

Problem: Sum of main diagonal of a square matrix.

Input: [[1,2,3],[4,5,6],[7,8,9]]

Output: 15

Hint: 1+5+9

Q103. Transpose Matrix

Problem: Print the transpose (rows become columns).

Input: [[1,2,3],[4,5,6]]

Output: 1 4\n2 5\n3 6

Hint: Swap indices

Section 9 — Strings (Q104–Q115)

String problems train you to think in characters and indices — closely related to arrays.

Q104. String Length

Problem: Print the length of a string without using length() (using a loop).

Input: "hello"

Output: 5

Hint: Try str.toCharArray()

Q105. Print Each Character

Problem: Print each character of a string on a new line.

Input: "Java"

Output: J\na\nv\na

Hint: Use charAt

Q106. Count Vowels

Problem: Count vowels in a string.

Input: "Hello World"

Output: 3

Hint: Check for a/e/i/o/u (both cases)

Q107. Count Consonants

Problem: Count consonants in a string.

Input: "Hello"

Output: 3

Q108. Reverse a String

Problem: Reverse a string.

Input: "Java"

Output: "avaJ"

Hint: Loop from end to start

Q109. Check Palindrome

Problem: Check if a string reads the same forward and backward.

Input: "madam"

Output: Palindrome

Hint: Compare with reversed

Q110. Count Words

Problem: Count how many words a sentence has.

Input: "Hello World Java"

Output: 3

Hint: Count spaces + 1

Q111. Capitalize Words

Problem: Capitalize the first letter of each word.

Input: "hello world"

Output: "Hello World"

Hint: Track previous-space state

Q112. Find First Non-Repeating Char

Problem: Find the first character that doesn't repeat.

Input: "aabbccdd"

Output: c

Hint: Count frequencies

Q113. Check Anagram

Problem: Check if two strings are anagrams (same letters, different order).

Input: "listen" "silent"

Output: Anagram

Hint: Sort or compare frequencies

Q114. Remove Spaces

Problem: Remove all spaces from a string.

Input: "Hello World"

Output: "HelloWorld"

Q115. Toggle Case

Problem: Convert uppercase to lowercase and vice versa.

Input: "Hello"

Output: "hELLO"

Hint: Check Character.isUpperCase

Section 10 — Methods (Q116–Q120)

Wrap reusable logic into methods. Practice taking parameters and returning values.

Q116. isEven Method

Problem: Write a method isEven(int n) that returns true if n is even.

Input: isEven(4)

Output: true

Q117. max Method

Problem: Write max(int a, int b) returning the larger of the two.

Input: max(5, 9)

Output: 9

Q118. Overloaded sum

Problem: Write 3 overloaded sum methods: (int,int), (int,int,int), (double,double).

Input: sum(2,3)

Output: 5

Hint: Same name, different params

Q119. printTable Method

Problem: Write a void method printTable(int n) that prints n's table 1-10.

Input: printTable(5)

Output: 5 10 15 ... 50

Q120. isPrime Method

Problem: Write isPrime(int n) returning true/false. Use it inside a loop to print primes 1-50.

Input: isPrime(7)

Output: true

Hint: Optimize using $i*i \leq n$

How to Approach These Problems

A simple loop you can apply to every problem in this set:

- Read the problem twice — make sure you understand input and expected output
- Dry-run on paper for a small input before touching the keyboard
- Write pseudocode in plain English — no Java syntax yet
- Convert pseudocode to Java line by line
- Test with the given input, then try 2-3 edge cases (0, negative, empty, large)
- Ask: "Is there a simpler or faster way?" Try the second pass

Edge Cases to Watch For

- Empty input — empty arrays, empty strings
- Single element — does your loop still work for $n=1$?
- Negative numbers — does your logic break?
- Zero — division by 0 will crash
- Very large input — does it overflow int? Switch to long if needed
- Duplicate values — does counting still work correctly?

Beginner Mistakes to Avoid

- Using `==` for String comparison instead of `.equals()`
- Off-by-one errors with array indices (last index is `length-1`, not `length`)
- Forgetting to update the loop variable inside a while loop (infinite loop!)
- Confusing `length()` (for Strings) with `length` (for arrays)
- Mixing `print()` and `println()` and wondering why the output looks wrong
- Not storing the original value before modifying it inside a loop

After You Finish These

Once you can solve most of these without help, you are ready for the next stage:

- Searching algorithms (Linear Search, Binary Search)
- Sorting algorithms (Bubble, Selection, Insertion)
- Recursion
- Two-pointer and sliding window problems
- LeetCode Easy and CodeChef Beginner problems

Key Takeaway: Don't aim to finish all 120 in a week. Aim to genuinely UNDERSTAND every problem you solve. Five problems deeply understood beat fifty solved by copy-paste. The goal isn't to check off questions — it's to rewire how you think.

Good luck! Come back to this list often — even after you've moved on, redoing these is a great warm-up before any DSA session.